

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Zoology

Paper: ZOB-101 — Systematics & Animal Diversity and Animal Form & Function
(Old Course 2009-10 Syllabus)

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Systematics & Animal Diversity) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) Differentiate between the following: [2 × 2 = 4]
- (i) *Petromyzon* and *Myxine*
 - (ii) Prototheria and Eutheria.
- (b) Define the following: [1 × 3 = 3]
- (i) Binomial Nomenclature
 - (ii) Species
 - (iii) Homology.
- (c) Write whether the following statements are True or False, giving reasons in two to three sentences: [2 × 2 = 4]
- (i) Honey bees belong to class Crustacea of phylum Arthropoda.
 - (ii) Water vascular system is found in *Asterias*.

Question 2. Answer the following parts: [6 + 6 = 12]

- (a) Discuss the distinguishing features of lizards and snakes with at least three examples of each group.
- (b) What is species concept? Discuss typological and biological species concept.

Question 3. Answer the following parts: [6 + 6 = 12]

- (a) Enumerate salient features of the animals belonging to Phylum Mollusca. Classify this phylum up to its different classes with their examples.

- (b) Describe flight adaptations in birds.

Question 4. Write notes on the following:

[2 × 6 = 12]

- (a) Hemichordates
(b) Dipnoi.

SECTION – B
(Animal Form & Function) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) Define the following: [1 × 3 = 3]
- (i) Food vacuole
 - (ii) Malpighian tubules
 - (iii) Microglia.
- (b) State whether the following statements are True or False giving reasons: [2 × 4 = 8]
- (i) Closed circulatory system is present in Earthworm.
 - (ii) Flame cells are present in vertebrates for excretion.
 - (iii) Bidder's canal is present in aves.
 - (iv) Axon terminals are involved in the formation of synapse.

Question 2. Answer the following parts:

[6 + 6 = 12]

- (a) With the help of a suitable diagram describe the function of mammalian kidney.
- (b) Explain the circulatory system of a cockroach.

Question 3. Answer the following parts:

[6 + 6 = 12]

- (a) Draw a labeled diagram of ommatidium and write its function.
- (b) Discuss different types of asexual reproduction in animals.

Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Structure and function of gills
- (b) Parthenogenesis
- (c) Phonoreception in fish.

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